Spiking Threshold Sweep Comparison Report: $\theta \in \{1.0, 0.8, 0.6, 0.4\}$ on MNIST SNN

ByzFL SNN Benchmark

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Contents

1 Introduction

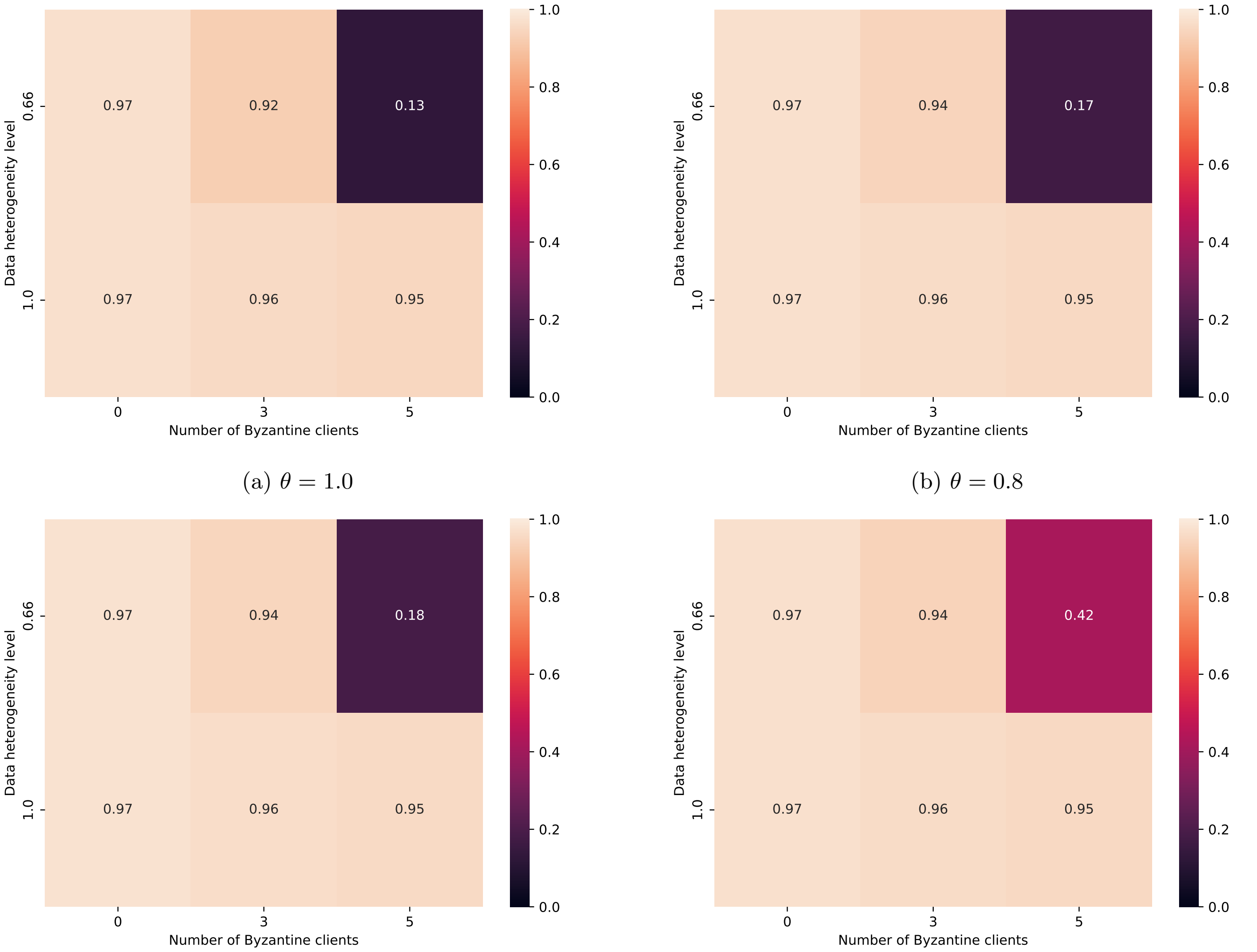
This report compiles and compares the robust aggregation sweep results for different values of the spiking threshold $\theta \in \{1.0, 0.8, 0.6, 0.4\}$. All models are SNNs trained with constant encoding ($T = 10$) and the Atan surrogate gradient ($\alpha = 1.2$). We compare the worst-case test accuracy across four robust aggregators (Centered Clipping, Geometric Median, Multi-Krum, and Trimmed Mean) under three different attacks:

1. Optimal A Little Is Enough (OptALIE)
2. Optimal Inner Product Manipulation (OptIPM)
3. Sign Flipping

Our objective is to verify whether varying the spiking threshold alters the gradient sparsity and consequently shifts the vulnerability window of the SNN.

2 Worst-Case Robustness (Across All Attacks)

The following figures show the worst-case test accuracy across all three attacks for the different threshold values.



(a) $\theta = 1.0$

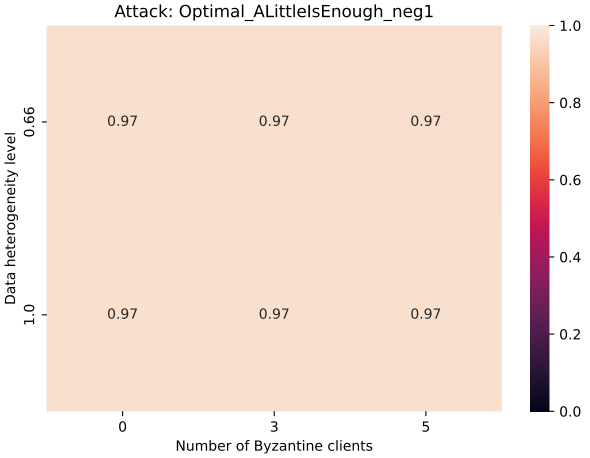
(b) $\theta = 0.8$

(c) $\theta = 0.6$

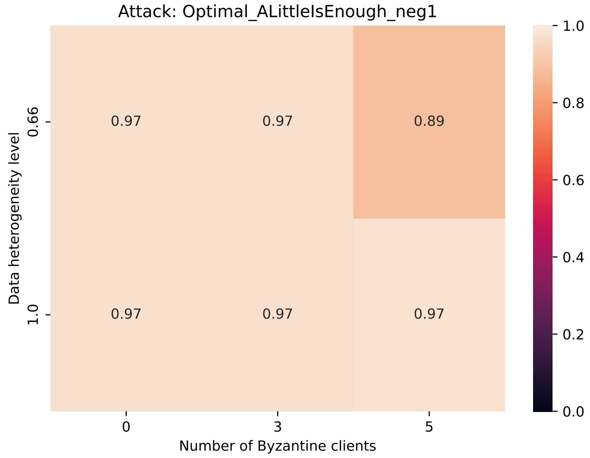
(d) $\theta = 0.4$

Figure 1: Worst-case test accuracy across all attacks (Worst-Case Heatmaps) for different spiking thresholds.

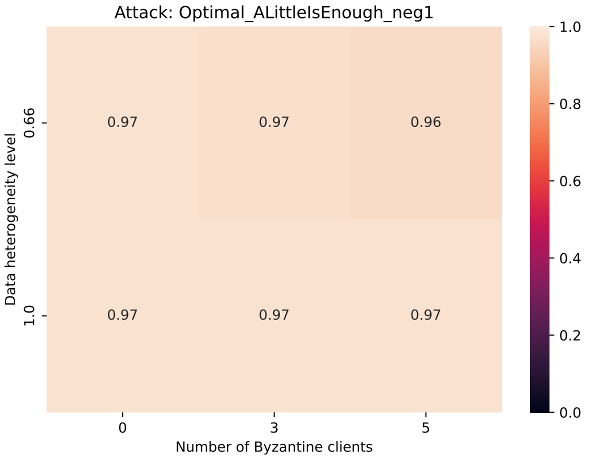
3 Robustness under Optimal ALIE



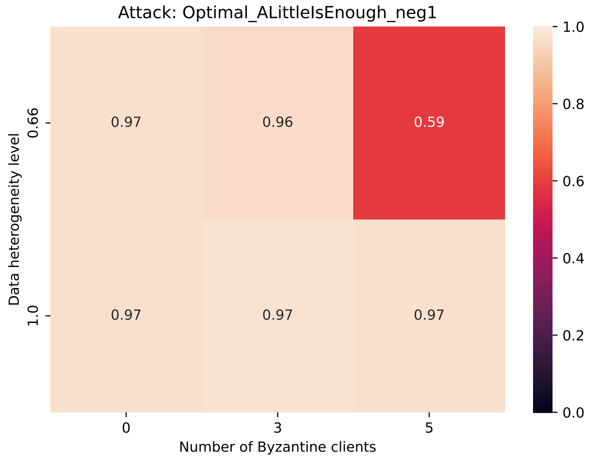
(a) $\theta = 1.0$



(b) $\theta = 0.8$



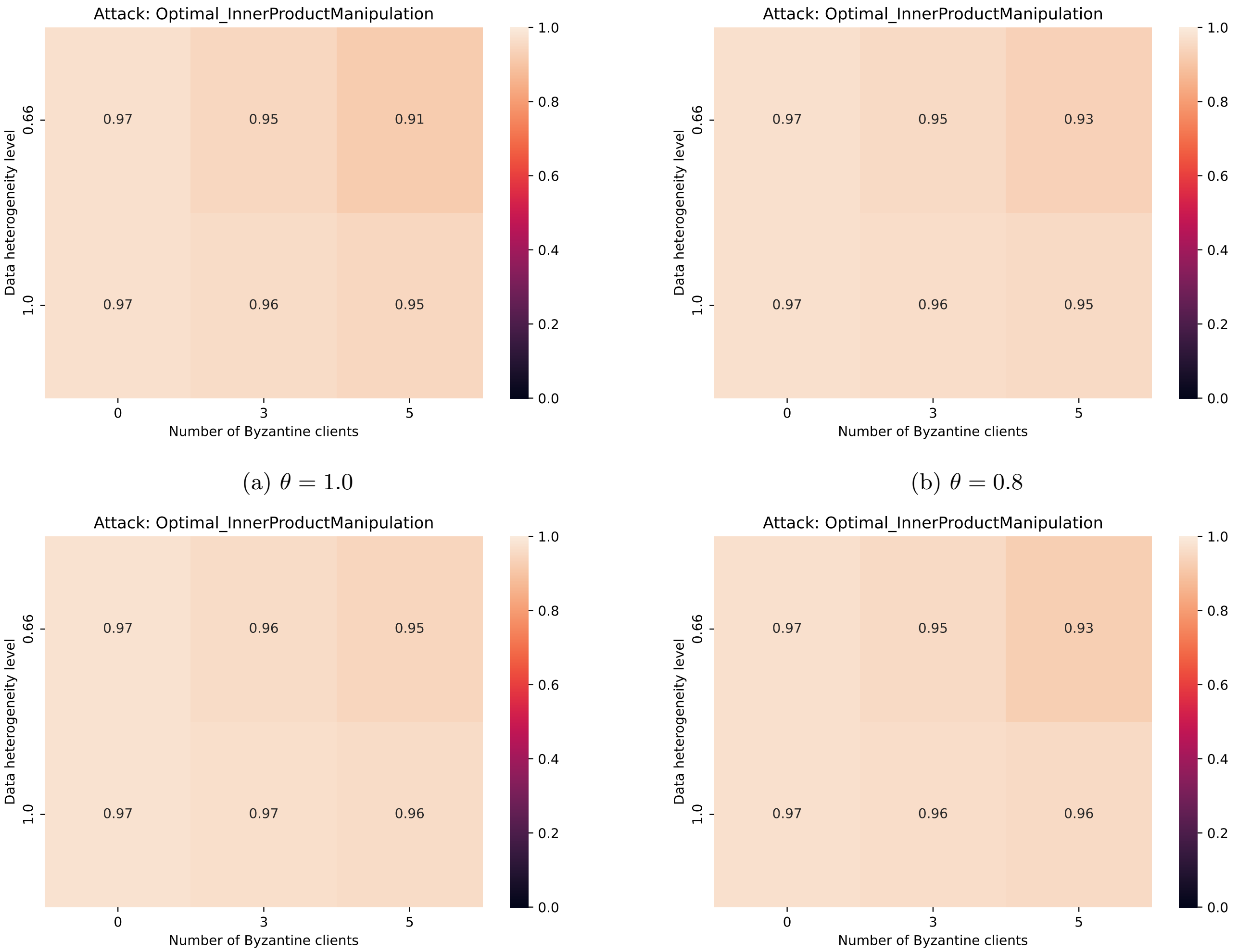
(c) $\theta = 0.6$



(d) $\theta = 0.4$

Figure 2: Test accuracy under Optimal A Little Is Enough (OptALIE) attack for different spiking thresholds.

4 Robustness under Optimal IPM



(a) $\theta = 1.0$

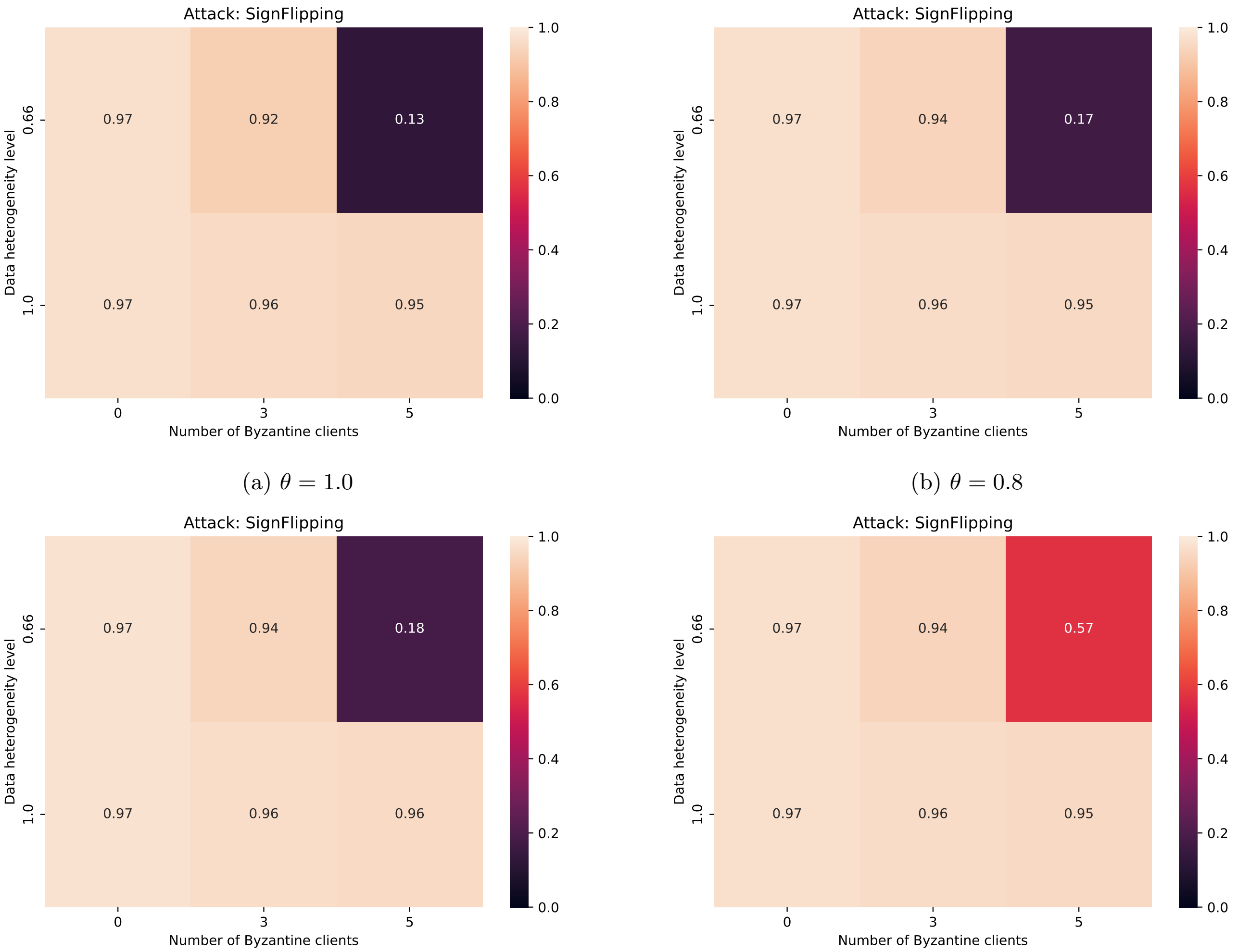
(b) $\theta = 0.8$

(c) $\theta = 0.6$

(d) $\theta = 0.4$

Figure 3: Test accuracy under Optimal Inner Product Manipulation (OptIPM) attack for different spiking thresholds.

5 Robustness under Sign Flipping



(a) $\theta = 1.0$

(b) $\theta = 0.8$

(c) $\theta = 0.6$

(d) $\theta = 0.4$

Figure 4: Test accuracy under Sign Flipping attack for different spiking thresholds.